ARUNJYOTI SARMA

Doctoral Research Scholar | Indian Institute of Technology (BHU), Varanasi

FPGA Systems Architect • Flexible Electronics & Micro-Sensors • Embedded Systems & Edge AI

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GitHub: github.com/arunjyotisarmarsece24-jpg/neuroquant-lob | **Nationality:** Indian

1. EXECUTIVE SUMMARY & RESEARCH PROFILE

Hardware Systems Engineer and Doctoral Scholar at the Department of Electronics Engineering, **Indian Institute of Technology (BHU) Varanasi**, specializing in flexible electronics, strain-sensing structural health monitoring (SHM), and ultra-low-latency FPGA digital architectures. Demonstrated track record of translating advanced multi-physics modeling and analog sensor conditioning into high-performance silicon: designed and physically closed a 322 MHz Spartan-7 FPGA deterministic limit-order-book engine (sub-microsecond latency, zero DSP, 82 mW power), pioneered flexible PVDF piezoelectric biomechanical energy harvesting (lead author, IEEE-ECBIOS 2026), and architected novel computing IP (USR-X gear-shift BNN/MAC register; VEGA RISC-V Processing-in-Memory cancer diagnostic accelerator for DVCon India 2025). Combines 8+ years of dedicated self-directed engineering R&D;, continuous national engineering excellence (four-time GATE qualifier in EC and EE: 2019, 2021, 2023, 2024), and premier doctoral research at India's foremost technical institutions.

2. EDUCATION & ACADEMIC APPOINTMENTS

Ph.D. in Electronics Engineering (Doctoral Scholar) — *IIT (BHU), Varanasi* *July 2024 – Present*

- **Doctoral Research:** Flexible electronics, thin-film strain-sensing arrays for structural health monitoring (SHM), piezoelectric nanogenerators (PENG), analog front-end readout circuits, and embedded digital telemetry.
- **Completed Coursework (54 Total Credits):** Signal Processing (EC-511), Information Theory & Coding (EC-512), Neural Networks (EC-523), Microprocessor Engineering & Applications (EC-521), Mobile Communication & Wireless Networks (EC-513), Switching Theory & Logic Design (EC-522).
- **Doctoral Fellowship:** Conferred full-time Institute Research Fellowship by the Ministry of Education (MoE), Government of India.

B.Tech. in Electronics & Electrical Engineering — *KIIT University, Bhubaneswar* *2012 – 2016*

- **Academic Standing:** CGPA: **7.51 / 10.0** (First Class with Distinction). Active Member, KIIT Robotics Society.
- **Core Domains:** Power Electronics, Microprocessors & Microcontrollers, Analog & Digital VLSI, Control Systems, Signals & Systems, C/C++.
- **Undergraduate Capstone:** Industrial Induction Motor Control via DTMF Telemetry; Autonomous Renewable Energy Generation Systems.

3. PROFESSIONAL & ADVANCED ENGINEERING RESEARCH EXPERIENCE

Independent Systems & Hardware R&D; Engineer — *Applied Research & Prototyping Laboratory* *June 2016 – June 2024*

- Executed an 8-year sustained research program in digital RTL acceleration, embedded architectures, and rigorous competitive engineering benchmarks; qualified national GATE examinations in four distinct cycles (2019, 2021, 2023, 2024) across EC and EE branches.
- **IP Architecture (USR-X):** Invented and validated a multi-mode gear-shift Universal Shift Register featuring barrel network stride shifting, LFSR PRPG/MISR logic built-in self-test (LBIST), and dual-mode compute agility (bit-serial MAC for MLPs and XNOR+popcount BNNs).
- **Embedded Systems & Firmware:** Developed custom firmware for ARM Cortex, ATmega, and PIC microcontrollers for industrial motor drives, acoustic sensing, and wireless telemetry.
- **Premier Institute Selection:** Shortlisted and defended technical research proposals across top IITs, culminating in selection for the Ph.D. program at IIT (BHU) Varanasi.

Academic & Technical Mentor (Electrical & Electronics Engineering) — *Engineering Foundations* *August 2017 – December 2023*

- Mentored aspiring engineering scholars in Microprocessors, Power Electronics, Circuit Theory, Signals & Systems, and Digital Logic Design.
- Guided hands-on lab sessions in microcontroller interfacing, circuit debugging, and algorithmic implementation in Python and MATLAB.

4. KEY SILICON IMPLEMENTATIONS & ENGINEERING SYSTEMS

NeuroQuant-LOB: Ultra-Low-Latency FPGA Limit Order Book & Inference Engine [*AMD Spartan-7 • Verilog • C++ • Python*]

- **Physical Silicon Verification:** Closed and deployed on physical Digilent Arty S7-25 (Spartan-7 XC7S25) on COM9 alongside an Arduino Nano hardware-in-the-loop (HIL) sidecar on COM10.
- **Timing & Silicon Footprint:** Achieved 322.06 MHz wire-speed timing closure (+0.894 ns positive slack) using **0 DSP slices** (100% fixed-point LUT arithmetic, consuming only 240 / 14,600 LUTs = 1.64% silicon footprint).
- **Ultra-Low Power:** Verified total on-chip dissipation of **82 milliwatts (0.082 W)** — yielding a 3,049x throughput/watt energy efficiency over enterprise Intel Xeon servers.
- **Deterministic SLA:** Achieved Median P50 latency = 1.230 µs and Tail P99 = 2.451 µs (exceeding institutional <10.0 µs SLA); walk-forward Deflated Sharpe Ratio DSR = 1.0000 across 500 permutations.
- **Software Suite:** Integrated automated verification harnesses, C++ pybind11 bindings, and real-time Flask/WebSocket mission control dashboard (http://localhost:5000).

PIM Edge AI Cancer Diagnostic Accelerator (DVCon India 2025) [*AMD Kintex-7 • VEGA RISC-V • Systolic Array*]

- Architected a Processing-in-Memory (PIM) AI inference accelerator integrated with India's indigenous VEGA RISC-V processor on a Genesys-2 (Kintex-7 XC7K325T) FPGA board.

- Embedded a Systolic Matrix Multiplication Engine (MME) directly into BRAM arrays to eliminate von Neumann memory bottlenecks; orchestrated DMA transfers via AXI4 memory-mapped interfaces.
- Executed quantized INT8 Convolutional Neural Network (CNN) / Vision Transformer (ViT) kernels for real-time histopathological slide classification with <20 ms inference latency and <6.8 W peak power.

USR-X: Secure, Self-Testing, Power-Gated Gear-Shift Shift Register IP [*Verilog HDL • Yosys • GTKWave • Python*]

- Designed and validated a computing register IP integrating segmented clock gating, multi-bit stride gear-shift barrel shifter, LFSR PRPG/MISR logic BIST, and stream scrambler scan chain security.
- Enabled dual-mode compute agility: operates as a bit-serial multiply-accumulate (MAC) core for MLPs/CNNs, an XNOR+popcount Binary Neural Network (BNN) accelerator, an FIR filter, and a stream cipher primitive.

5. PEER-REVIEWED PUBLICATIONS & INTELLECTUAL PROPERTY

- **Refereed International Conference Paper (Lead Author & Presenter):** Arunjyoti Sarma, Kuntal Kumar Das, Dr. L. Oppili Prasad, Dr. Ashutosh Kumar Dubey, *"Piezoelectric Nanogenerator-Based Energy Harvesting System for Low-Power Bioelectronics,"* **Proceedings of the 2026 IEEE 8th Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (IEEE-ECBIOS 2026)**, Tainan, Taiwan, October 2026. (Joint Research: Dept. of Electronics Engg. & Dept. of Ceramic Engg., IIT (BHU) Varanasi).
- **Patented Architecture & Invention Disclosure (Sole Inventor):** Arunjyoti Sarma, *"A Reconfigurable, Power-Gated Universal Shift Register with Multi-Stride Barrel Shifting and Built-In Self-Test for Bit-Serial Neural Acceleration and Cryptographic Primitives (USR-X),"* Indian & International PCT Patent Specification, Ref: USR-X/IIT-BHU/2025. (RTL verification & silicon simulation closed).
- **National Conference Technical Monograph (Lead Designer):** Arunjyoti Sarma, Swati Gupta, Updesh Kumar (Team AstraVortex, IIT (BHU) Varanasi), *"Processing-in-Memory (PIM) Hardware Accelerator Integrated with Indigenous VEGA RISC-V Processor for Real-Time Edge Cancer Diagnostics,"* **DVCon India 2025 Design Contest Technical Monograph**, Bangalore, India.
- **Competitive Research Fellowship Proposal (Principal Investigator):** Arunjyoti Sarma, *"Sub-Microsecond Deterministic Limit Order Book Engine and Neuromorphic FPGA Acceleration on Sub-Watt Silicon,"* **NVIDIA Graduate Fellowship Research Proposal & Bitter Lesson Technical Manifesto** (Certified Institute Nomination Dossier, IIT (BHU) Varanasi).
- **Journal Manuscript in Preparation (Targeted Submission):** Arunjyoti Sarma, Dr. L. Oppili Prasad, *"Multi-Physical Strain Dynamics and Electromechanical Optimization of Serpentine Interconnects on Elastomeric Substrates for Continuous Structural Health Monitoring,"* Manuscript in preparation for submission to **IEEE Sensors Journal**.

6. NATIONAL HONORS, FELLOWSHIPS & ACADEMIC DISTINCTIONS

- **Ministry of Education (MoE) Institute Doctoral Research Fellowship (2024 – Present):** Conferred full-time Institute Research Fellowship by the Ministry of Education, Government of India, to pursue advanced doctoral research in the Department of Electronics Engineering, Indian Institute of Technology (BHU) Varanasi.
- **Dual-Discipline National Engineering Mastery (GATE — EC & EE):** Four-time national qualifier in the Graduate Aptitude Test in Engineering across both Electronics & Communication Engineering (EC) and Electrical Engineering (EE) — **GATE 2024, GATE 2023, GATE 2021, and GATE 2019**; demonstrates continuous, verified theoretical and applied command across solid-state physics, electromagnetics, signals, communication, and digital VLSI among top percentiles of engineers nationwide.
- **DVCon India 2025 National Hardware Design Contest (Stage 2 Finalist & Lead Architect):** Selected to Stage 2 of the premier international Design and Verification Conference (DVCon India 2025) Design Contest as Lead Architect of Team AstraVortex (IIT BHU); architected an indigenous VEGA RISC-V integrated Processing-in-Memory (PIM) accelerator on AMD Kintex-7 FPGA for real-time edge histopathology classification (<20 ms latency, 6.8 W peak power).
- **Institutional Research Nomination — NVIDIA Graduate Fellowship (2026):** Nominated by the Department of Electronics Engineering, IIT (BHU) Varanasi, for the international NVIDIA Graduate Fellowship, proposing a 322.06 MHz Spartan-7 FPGA deterministic limit order book engine with 0 DSP slices and 82 mW power dissipation, validating Sutton's 'Bitter Lesson' at microsecond line-rate.

7. TECHNICAL SKILLS & CORE COMPETENCIES

- **FPGA & Digital Hardware Design:** Verilog HDL, SystemVerilog, AMD Vivado Design Suite, Spartan-7 / Artix-7 / Kintex-7, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Fixed-Point DSP-Free Arithmetic, RTL Synthesis, Yosys, Verilator, GTKWave, ModelSim, Hardware-in-the-Loop (HIL) Emulation.
- **Electronic Design Automation (EDA) & Simulation:** Cadence Innovus (P&R;, CTS), Cadence Genus (Synthesis), RTL-to-GDSII flow, COMSOL Multiphysics (Solid Mechanics, Electrostatics, Piezoelectric Devices, AC/DC), Tina-TI, LTspice, Proteus, Altium / KiCad PCB Layout.
- **Analog, Sensor & Biomedical Hardware:** Piezoelectric Nanogenerators (PVDF/PENG), Flexible Strain Sensing, Analog Front-End (AFE) Design, Instrumentation Amplifiers, Precision Bridge Readouts, MT3608 DC-DC Converters, Micro-Fabrication & Hot-Press Polymer Processing.
- **Programming & Embedded Firmware:** C, C++ (C++17/20, pybind11), Python (NumPy, SciPy, Pandas, Scikit-learn, PyTorch), MATLAB / Simulink, Embedded C, Microchip PIC, ATmega (Arduino), ARM Cortex-M, ESP32, Raspberry Pi, Linux Shell / PowerShell.
- **Architectures, Protocols & Web Telemetry:** RISC-V ISA (VEGA indigenous core), Processing-in-Memory (PIM), UART, SPI, I2C, AXI4-Lite/AXI4, FTDI High-Speed Serial, TCP/IP, WebSockets, Flask, Docker, Git, LaTeX, Signal Processing, Information Theory, Neural Networks, Real-Time Systems.